

IRIS-CAND-03

Canonical PostGIS Pilot Schema

Timebox	5 hours	Project basis	IRIS-003, IRIS-007
Primary stack	Python + PostgreSQL/PostGIS	Submission	Git repository / zip + run instructions

Objective

Design and implement the minimum reproducible PostgreSQL/PostGIS schema required by the IRIS pilot.

Scenario

The pilot needs staging and core schemas with country-aware keys, geometry contracts, source metadata, and enough constraints to support BESS and peatland inference.

Technical task

- Create SQL migrations for `iris_staging` and `iris_core`.
- Create minimal parcel, substation, peatland, screening_layer, source_run, and evidence tables.
- Use non-null `country_code` and country-scoped unique keys.
- Choose appropriate geometry types/SRIDs for storage and document the CRS policy.
- Add spatial and relational indexes needed for the two pilot verticals.
- Provide a reset/rebuild command and seed fixtures.

Starter fixture / contract

Required canonical names:
`geom`, `country_code`, `region_code`, `source_id`, `source_date`, `created_at`
Avoid a 'geometry' column name in the core contract.

Required deliverables

- SQL migrations
- Seed SQL or Python loader
- Schema diagram (text or Mermaid accepted)
- Verification queries

Acceptance criteria

1. Schema rebuilds from empty DB without manual intervention.
2. All required pilot query fields exist and match canonical names.
3. `country_code` cannot be null.
4. A sample spatial round trip and index usage are demonstrated.

Engineering constraints

- Use Python 3.12+ and PostgreSQL 16+ with PostGIS 3.4+ unless the assignment explicitly narrows the stack.
- Use a canonical geometry column named **geom**. Never silently invent missing geospatial attributes.
- Every persisted business entity must carry a non-null **country_code**; identifiers and joins must be country-scoped.
- Treat data completeness, CRS, units, source date, and uncertainty as explicit data contracts.
- Prefer reproducible CLI commands and SQL migrations over notebook-only work.
- No external production credentials are required. Use local fixtures/mocks where a source would otherwise be remote.
- Your submission must run locally from a clean checkout using the documented commands.

Evaluation rubric

Criterion	Weight
DDL correctness	30%
Key/constraint design	25%
Spatial design	20%
Reproducibility	15%
Documentation	10%

Submission expectations

- README: setup, assumptions, commands, and architecture choices.
- Automated tests covering the highest-risk correctness conditions.
- A small deterministic fixture set committed with the submission.
- No production secrets, paid APIs, or inaccessible proprietary data.
- Document any deliberate simplification and how it would evolve for production.

Project context

This exercise is derived from the Project IRIS delivery playbook. The pilot prioritizes a one-region vertical slice, country-aware reusable adapters, PostgreSQL/PostGIS, geometry correctness, controlled promotion, explainable spatial screening, and two-page prospecting outputs. The challenge intentionally tests production-minded engineering without requiring access to private IRIS infrastructure.

Reference uncertainty wording used by the project:

Preliminary prospecting material. Figures, eco-point estimates and site suitability are indicative and based on available source data and commercial screening assumptions. The 8 eco-points/m2 factor is the current commercial baseline, not certified compensation. Ownership, planning, grid capacity, environmental eligibility and transferability remain subject to project-specific verification. No permit, reservation or construction readiness is represented.